

Revenue & Operations Optimization

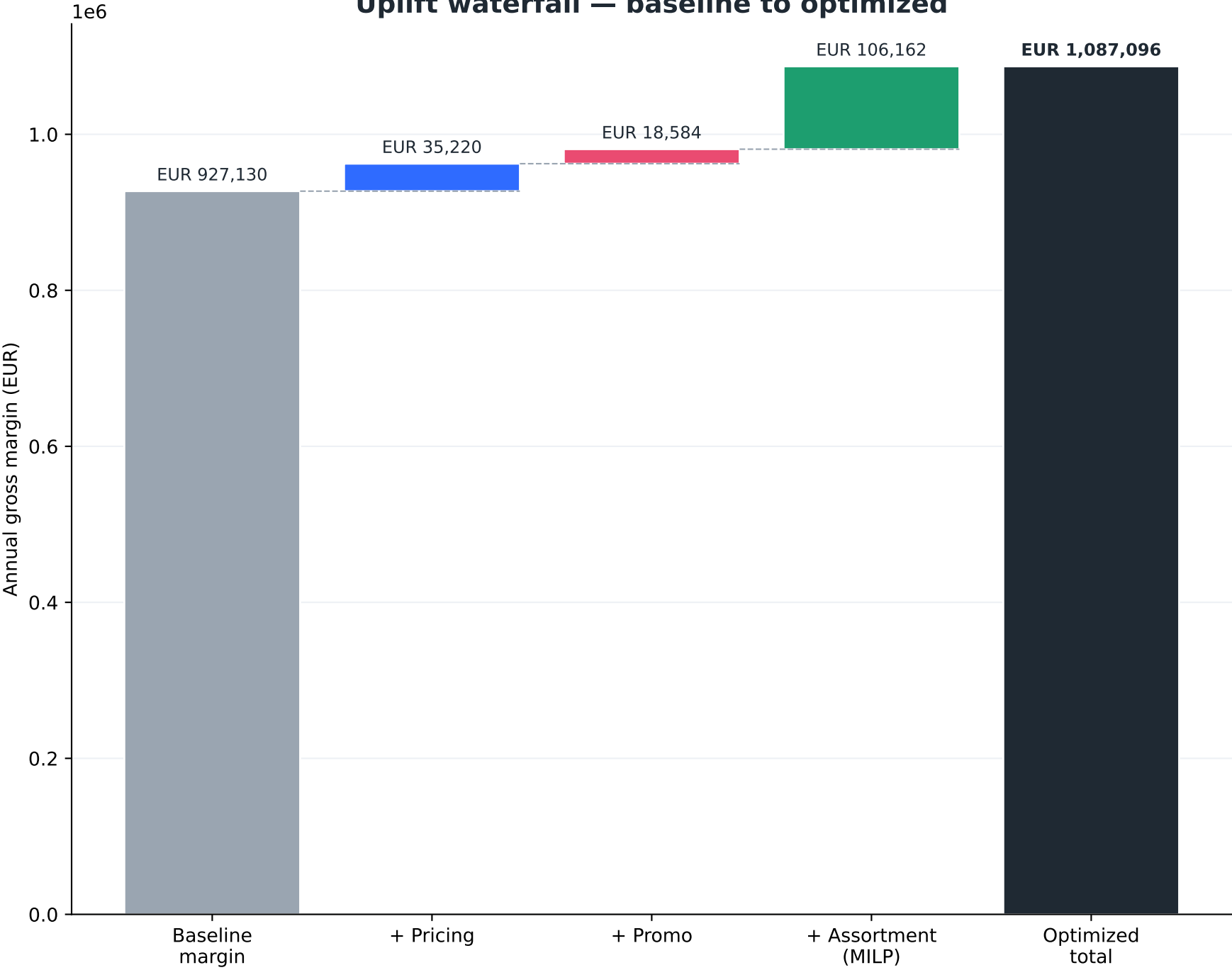
Executive Review

Total expected uplift EUR 159,966 / year

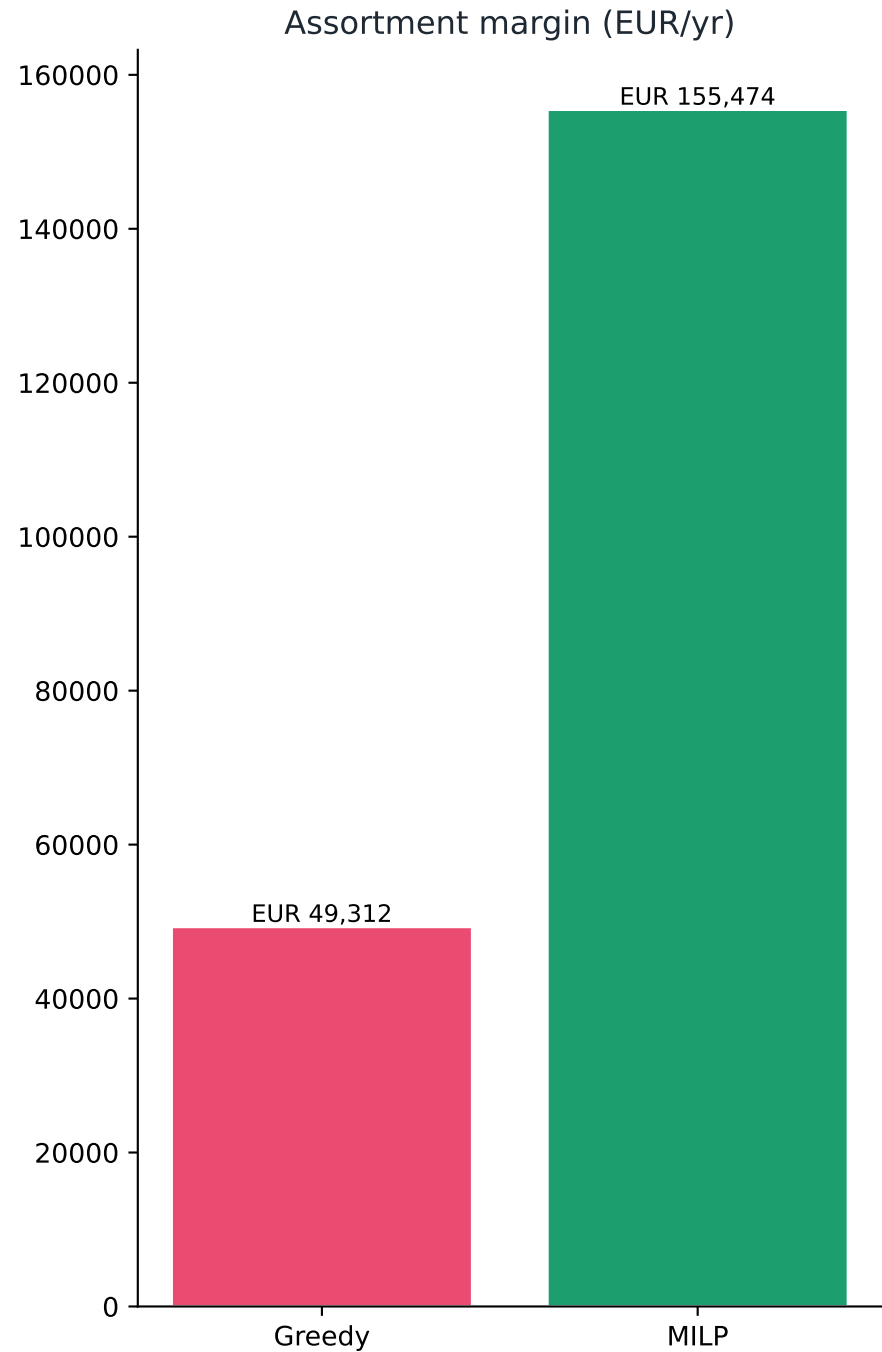
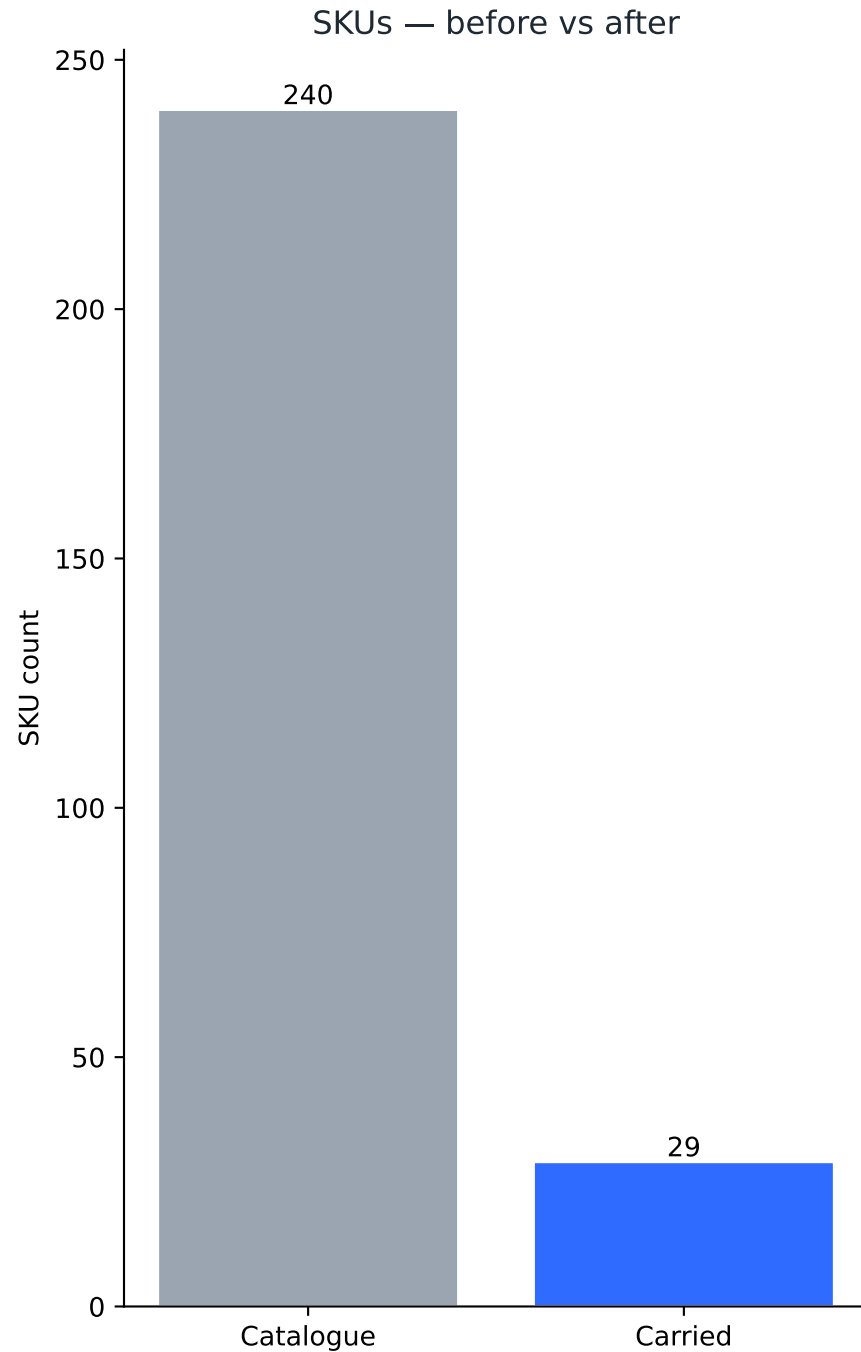
AI demand/elasticity/decline models feeding a MILP + newsvendor + pricing + promo optimization core

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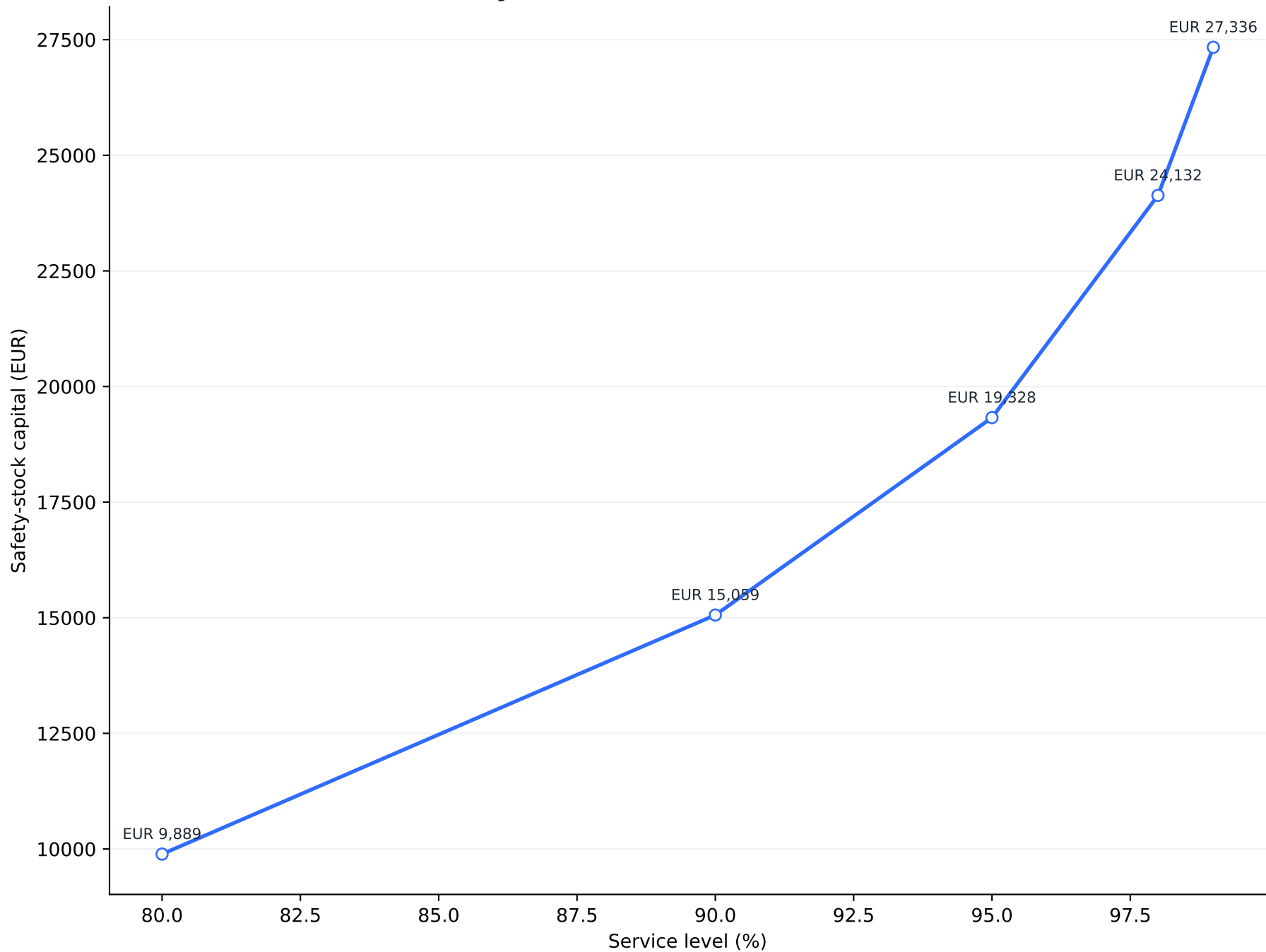
Uplift waterfall — baseline to optimized



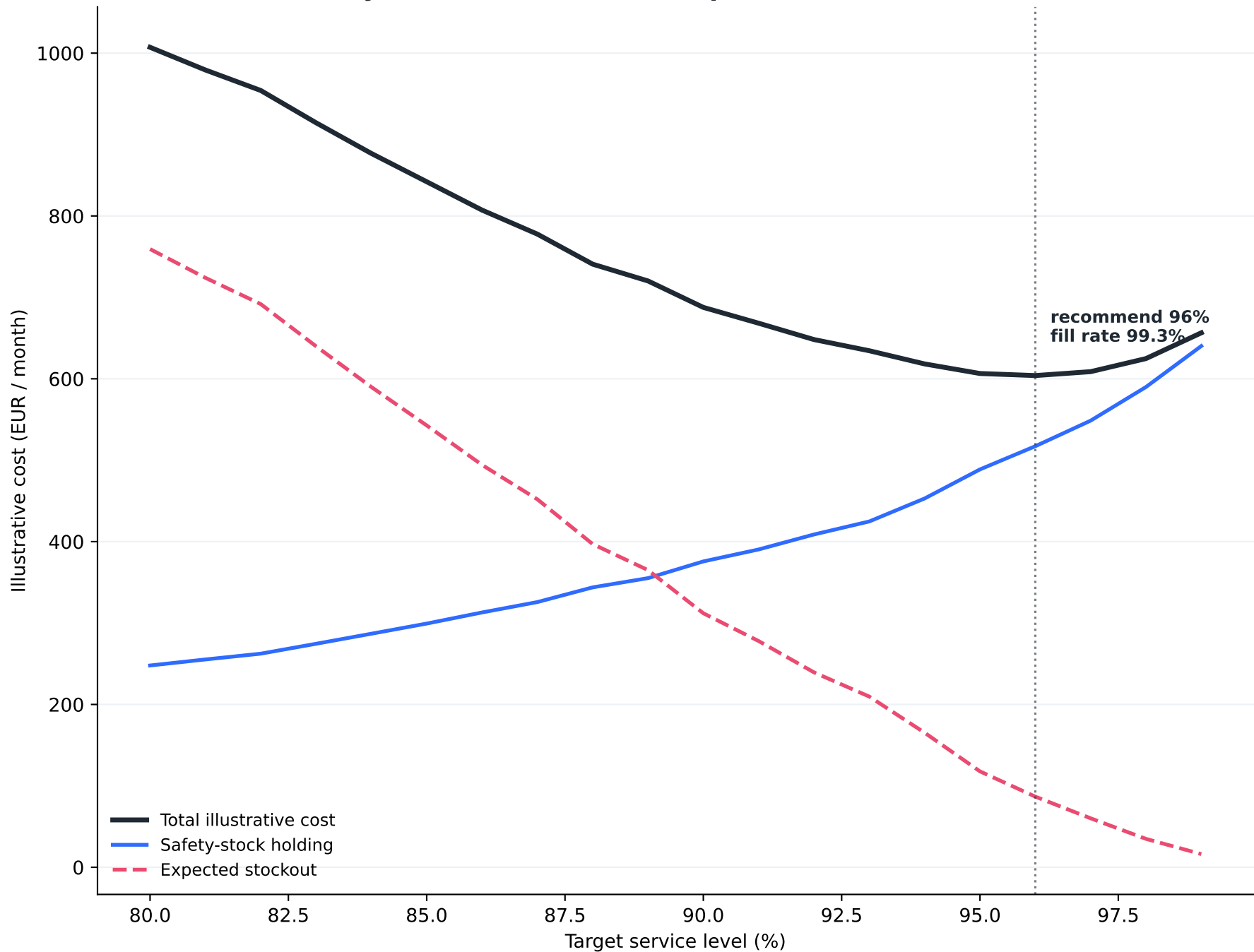
Assortment before/after — MILP earns EUR 106,162/yr over greedy



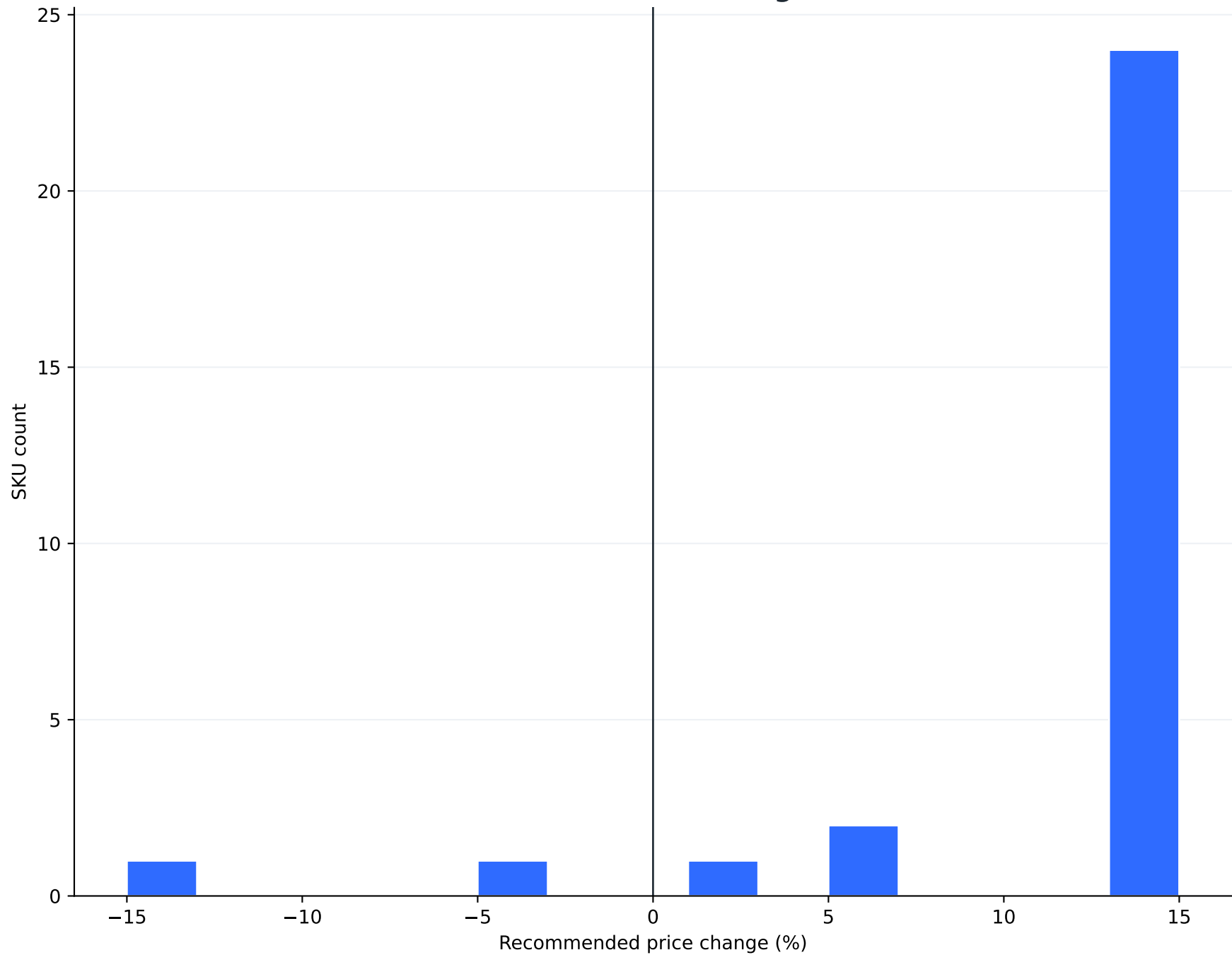
Inventory service-vs-cost frontier — S0006



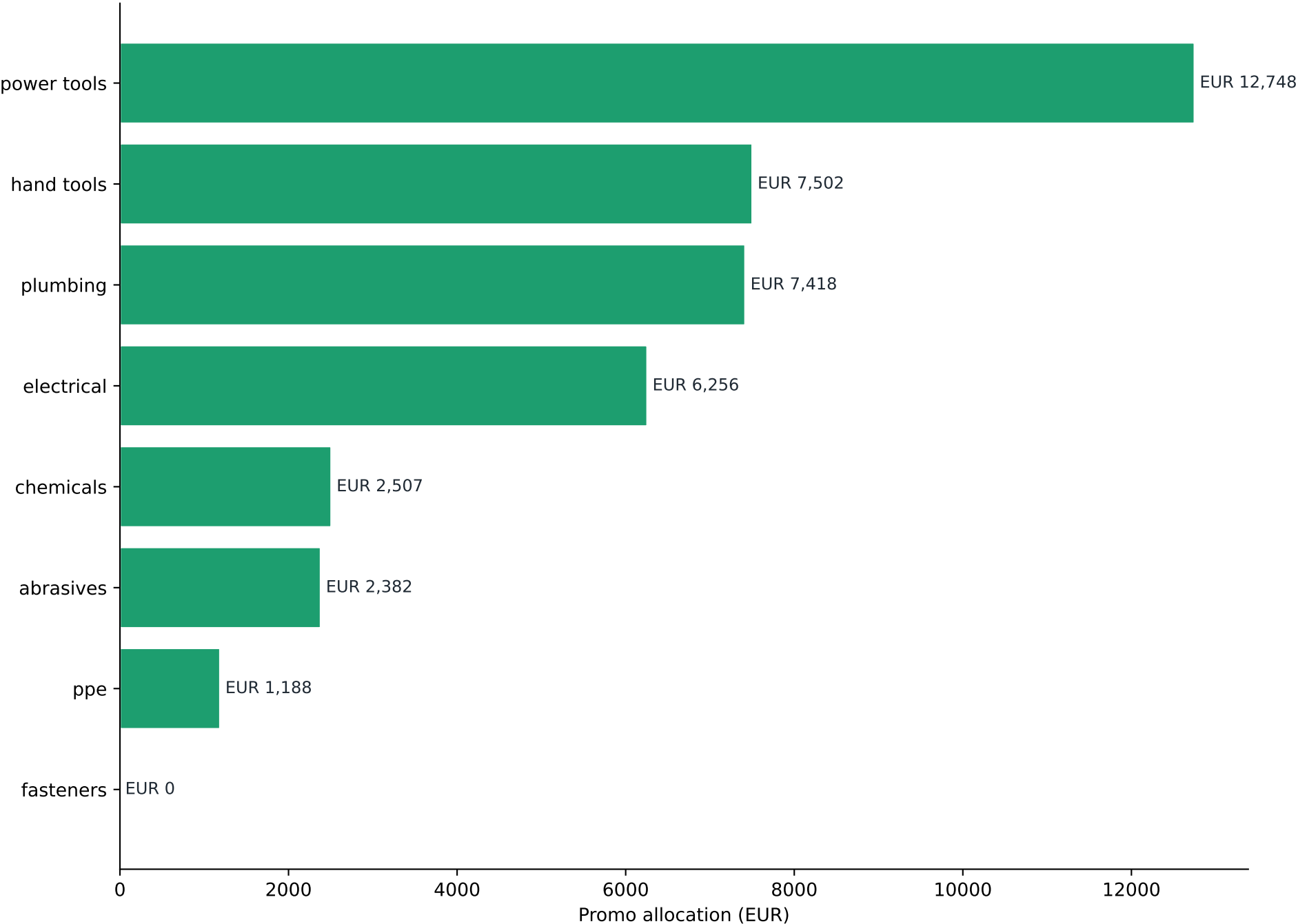
Forecast-uncertainty service-level curve (empirical 2.28σ vs Gaussian 1.75σ at reco)



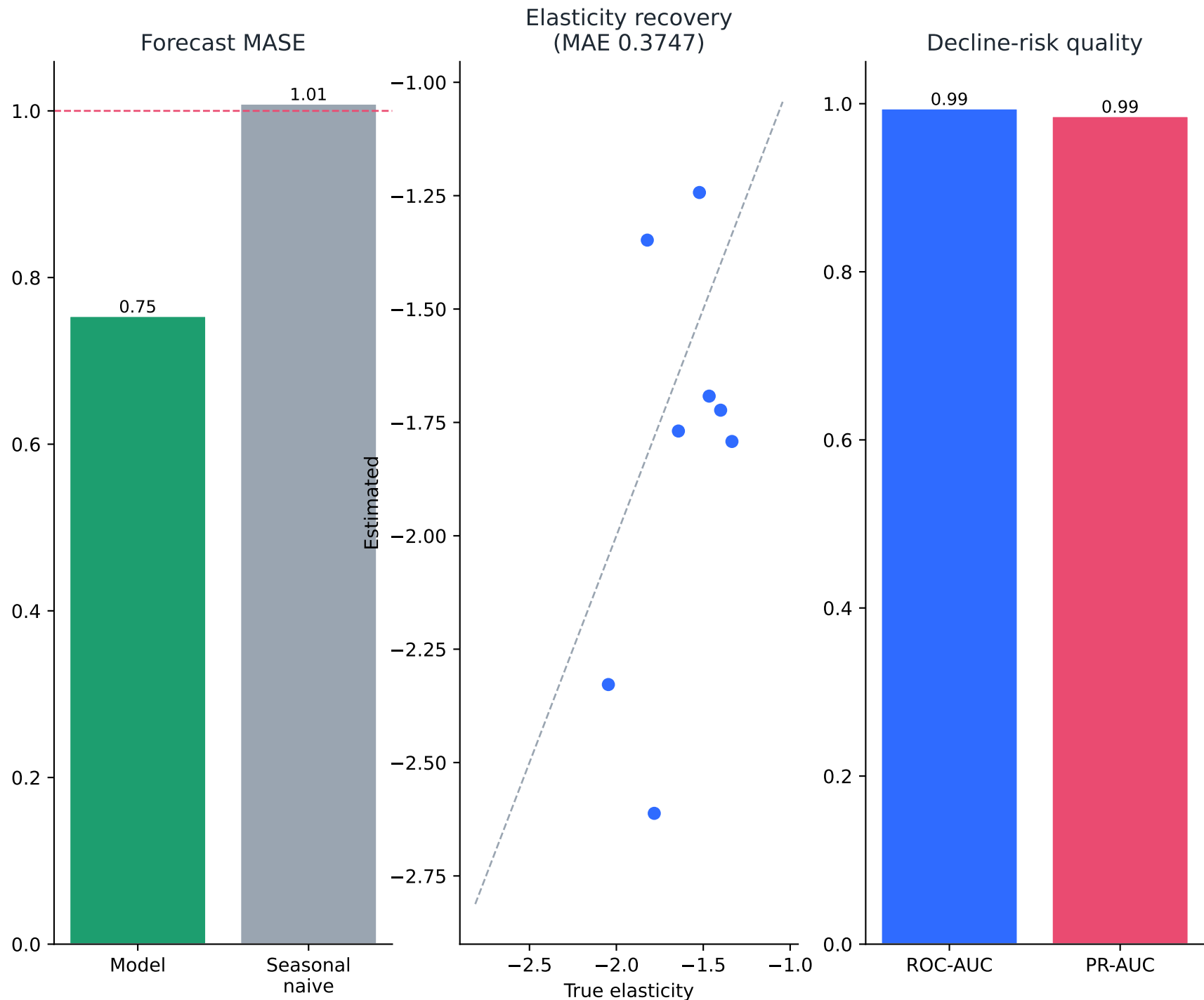
Price-move distribution (within guardrail band)



Promo allocation — EUR 18,584 incremental (+EUR 955 vs even split)



Model quality (honest, held-out)



Recommended actions

1. Assortment: carry 29 of 240 SKUs within EUR 60,000 working capital (EUR 56,152 used). The MILP earns EUR 106,162/yr over the GMROI heuristic; 71 of the dropped SKUs are flagged declining.
2. Price: raise S0006 by +15.0% (EUR 229.83 -> 264.30) -> EUR 15,838/yr [estimated elasticity -1.69, elastic].
3. Price: raise S0240 by +15.0% (EUR 150.20 -> 172.73) -> EUR 6,586/yr [estimated elasticity -1.69, elastic].
4. Price: raise S0031 by +15.0% (EUR 226.41 -> 260.37) -> EUR 4,125/yr [estimated elasticity -1.69, elastic].
5. Promo: split EUR 40,000 as power tools EUR 12,748, hand tools EUR 7,502, plumbing EUR 7,418, electrical EUR 6,256 -> EUR 18,584 incremental margin (EUR 955 better than an even split).
6. Inventory: set order-up-to & reorder points on 29 carried SKUs from the demand forecast (avg safety stock 22.7 units).
7. Service level: the forecast-residual curve recommends a 96% target (empirical 2.28 sigma vs Gaussian 1.75) -> ~99.3% expected unit fill rate at EUR 604/mo illustrative cost; assuming a bell curve would under-stock the fat-tailed forecast error.
8. Risk: 80 SKUs flagged declining (prob ≥ 0.5); 9 are still carried but demand-haircut. Decline classifier test ROC-AUC 0.9943.
9. Forecast quality: demand MASE 0.7538 vs seasonal-naive 1.0086 (beats naive); elasticity recovered to EUR-free MAE 0.3747.
10. Bottom line: total expected uplift EUR 159,966/yr (pricing 35,220 + promo 18,584 + assortment 106,162).